Phys 101 Worksheet

Comparing Model and Experiment

Name: _____ **Student #:** _____ **Student Email #:** _____

1. Suppose there's a theory "Dogs cannot be born with naturally blue fur."Can this theory be proven to be true? **Explain** why or why not.

 This theory **cannot be proven to be true**. It can only be disproved once a dog with naturally blue fir is discovered.

2. Suppose you're a butterfly researcher. You find that rarer butterflies appear to be faster, and flutter their wings at a higher rate than more common butterflies. You invent a new unit, a "Whimsy" (denoted as W), which is the **product** the maximum velocity of a given species of butterfly, v, and the frequency at which it flaps its wings f.

 (a) Explicitly write out the formula to calculate Whimsiness. What are it's dimensions?

 $W = vf$**. It has dimenions of:** $\frac{L}{T^2}$

 (b) Suppose now you want to create a formula that could predict the most whimsical butterfly any given butterfly catching machine could catch knowing only its mass, m and the maximum force it can swing with F_s. Write a possible solution (within a constant) that could be a solution to the problem.

 $W = k\frac{F_s}{m}$

 (c) Does the formula you've devised make sense? Explain wy or why not

 It makes sense because the formula is dimensionally sound. As well, a bigger machine would be harder to carry and thus would not be able to catch as whimsical butterflies which will be faster and thus more difficult to catch.